# marketTon

## Google Ads API - Tool Design Documentation

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|---|---|
| Tool name | marketTon |
| Website | https://felixsystems.com.br/ (privacy policy: https://felixsystems.com.br/privacidade.html) |
| Developer / operator | Everton Felix (Felix Systems), Brazil |
| Contact | sysefo.contato@gmail.com |
| Google Cloud project | MarketTon (project number 662521647054) |
| Access level requested | Basic Access |
| API used | Google Ads API (REST) - read-only, Keyword Planner metrics |
| Document date | 21 October 2026 |

## 1. Summary

marketTon is a private, internal market-research tool built and operated by a single person: an affiliate marketer who promotes products from online marketplaces (Mercado Livre and Shopee) in Brazil. The tool helps the operator decide **which products are worth promoting** by combining marketplace data (price, sales, seller reputation, commission) with search-demand data.

The tool uses the Google Ads API for exactly one purpose: to read **Keyword Planner historical metrics** (average monthly searches, competition level and estimated top-of-page bid/CPC) for keywords related to products the operator is evaluating. It does **not** create, edit, pause or delete campaigns, ads, ad groups, keywords, budgets or any other account object, and it does not read reporting data about the account's advertising performance.

Why Basic Access: the tool needs to call the Keyword Planner methods against a production account, which Google's access-level documentation associates with Basic Access. Expected usage is a few requests per day, far below the Basic Access limit.

## 2. Who uses the tool

- **Users:** only the operator (a single person). There are no external users, customers, agencies or sub-accounts.
- **Account:** the operator connects his **own** Google Ads account. The tool never accesses accounts belonging to third parties.
- **Distribution:** the tool is not sold, licensed or offered as a service today. If it is ever opened to other users, each user would connect only their own account and the privacy policy would be updated beforehand.

## 3. What the tool does with Google Ads data

The operator opens the page **Oportunidades de Busca** (Search Opportunities) and types or selects keywords. For each keyword the tool shows:

- average monthly searches (Brazil, Portuguese),
- competition level (low / medium / high), and

- estimated CPC in the account's currency (derived from the top-of-page bid range).

These values are shown next to the keyword's Google Trends trend (up/down and % change over 30 days), so the operator can find products with **rising demand and low advertising cost**. The result is a ranked 'Top opportunities' list. Nothing is written back to Google Ads.

## 4. Architecture and data flow

| Component | Technology | Role |
|---|---|---|
| Web interface | React single-page app | Operator UI (login required). Pages: Products, Portfolio, Search Opportunities, Integrations. |
| API server | Python / FastAPI | Authenticated REST API. Holds the Google Ads client and all credentials. |
| Database | PostgreSQL | Stores the Google Ads credentials of the operator's own account (server side only). |
| Google Ads API | REST over HTTPS | Called only from the API server, never from the browser. |

### Request flow:

1. The operator logs in to the marketTon interface (username/password, JWT session).
2. The interface calls the marketTon API: `GET /trends/opportunities?keywords=...`
3. The API server exchanges the stored refresh token for an access token (Google OAuth 2.0) and calls the Google Ads API.
4. The response is normalised, combined with Google Trends data, ranked, and returned to the interface.
5. Results are kept in an in-memory cache for 6 hours (to avoid repeated calls) and are **not** written to the database.

## 5. Google Ads API usage

### Authentication

- OAuth 2.0 with the scope `https://www.googleapis.com/auth/adwords`, using the operator's own OAuth client (Google Cloud project MarketTon). The OAuth consent screen is published, and the app's brand has been verified by Google.
- A refresh token is obtained once by the operator and stored on the server. It is write-only in the interface (never displayed again after saving).
- Requests carry the developer token and, when applicable, the manager account ID in the `login-customer-id` header.

### Methods called (all read-only)

| Method | Purpose | Frequency |
|---|---|---|
| customers.generateKeywordHistoricalMetrics (KeywordPlanIdeaService) | Average monthly searches, competition, competition index and top-of-page bid for up to 15 keywords per request. Parameters: geoTargetConstants/2076 (Brazil), languageConstants/1014 (Portuguese), network GOOGLE_SEARCH. | On operator action; results cached 6 h |

| Method | Purpose | Frequency |
|---|---|---|
| customers.googleAds:search (GoogleAdsService) | GAQL `SELECT customer.currency_code FROM customer LIMIT 1` - reads the account currency so the CPC is labelled correctly, and powers the 'Test connection' button. | On 'Test connection' and with each metrics request |

### Not used

- No mutate operations of any kind (no campaigns, ads, keywords, budgets, conversions, audiences, user lists or billing).
- No access to customer lists, conversion data or advertising performance reports.
- No Keyword Plan creation (the tool uses historical metrics only, without creating plans).

### Expected volume

The tool is operated by one person and queries on demand. Typical use is 1 to 10 requests per day (each request carries up to 15 keywords), i.e. far below 15,000 operations/day, and results are cached for 6 hours.

## 6. Screenshots

The screenshots below are from the running application. Credential fields never display saved secrets.

### 6.1 Integrations page - Google Ads card

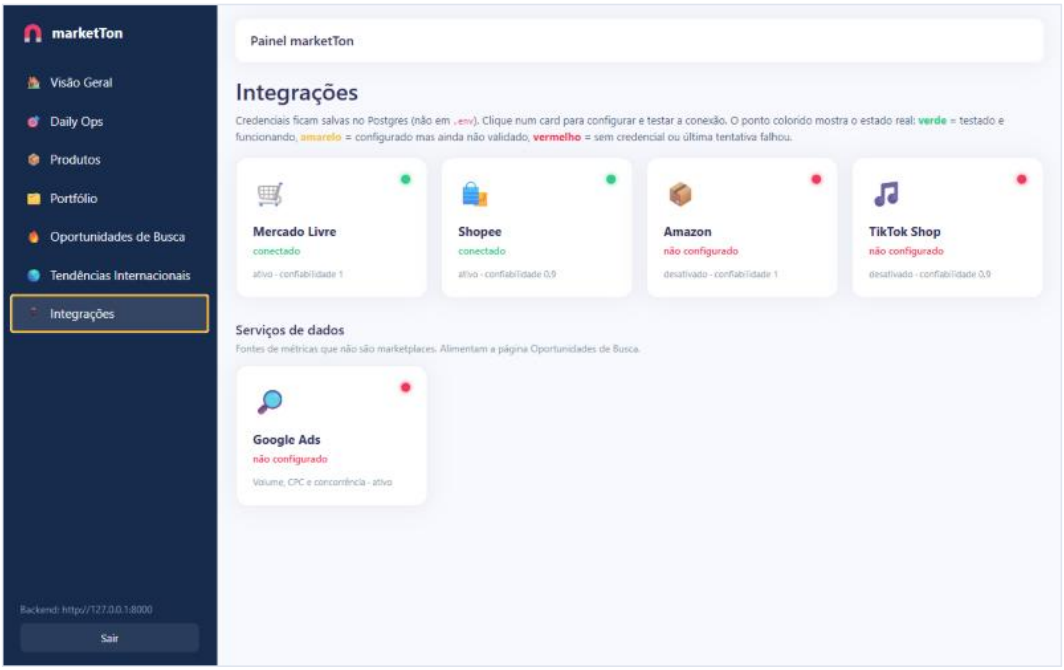



Figure 1. 'Serviços de dados' section on the Integrations page. The Google Ads card shows the connection status (traffic light: green only after a successful real test call).

## 6.2 Credentials form

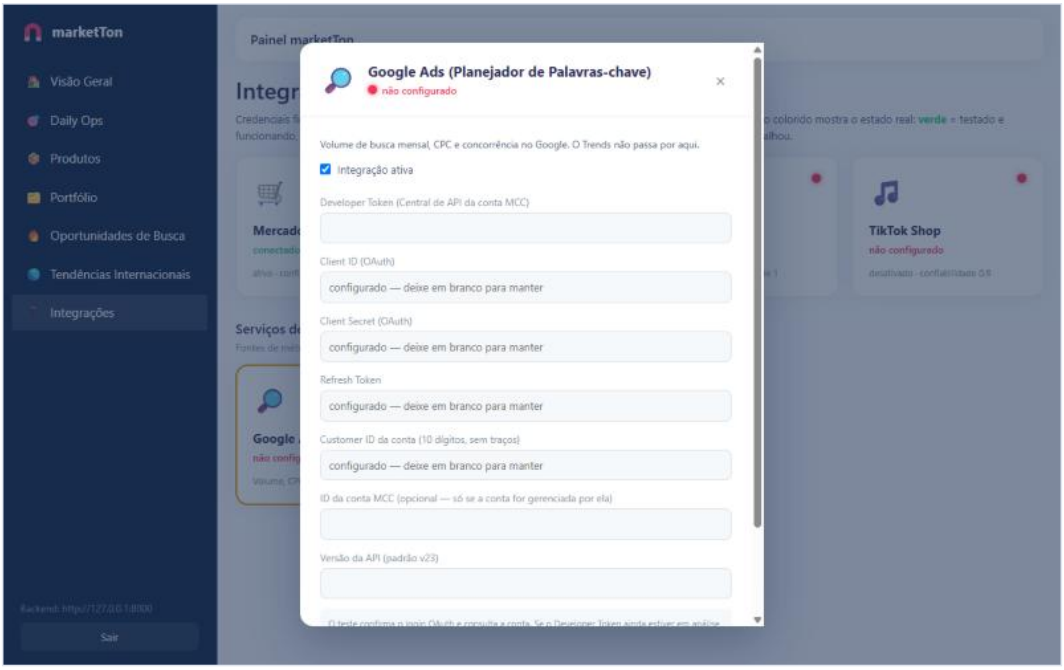



Figure 2. Credential form. Secret fields are never returned by the API; once saved they appear only as 'configurado' (configured). The 'Testar conexão' (Test connection) button performs the OAuth login and a read-only account query.

## 6.3 Search Opportunities - real data

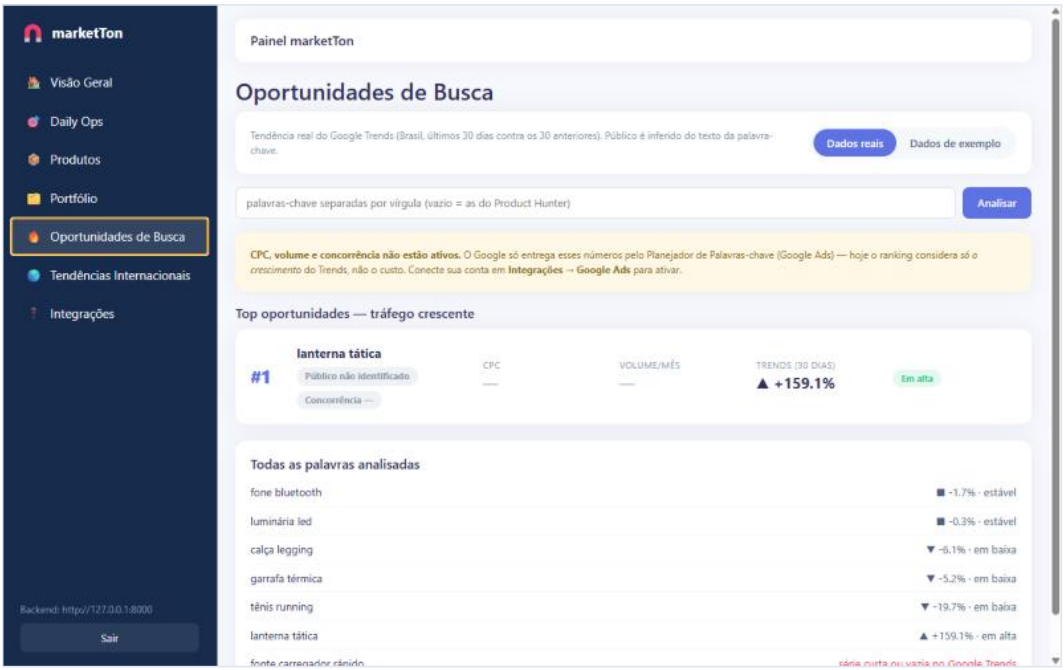



Figure 3. Search Opportunities page with live Google Trends data. When Keyword Planner metrics are not yet available, the interface says so and shows a dash instead of estimating values.

### 6.4 Search Opportunities - layout with Keyword Planner metrics

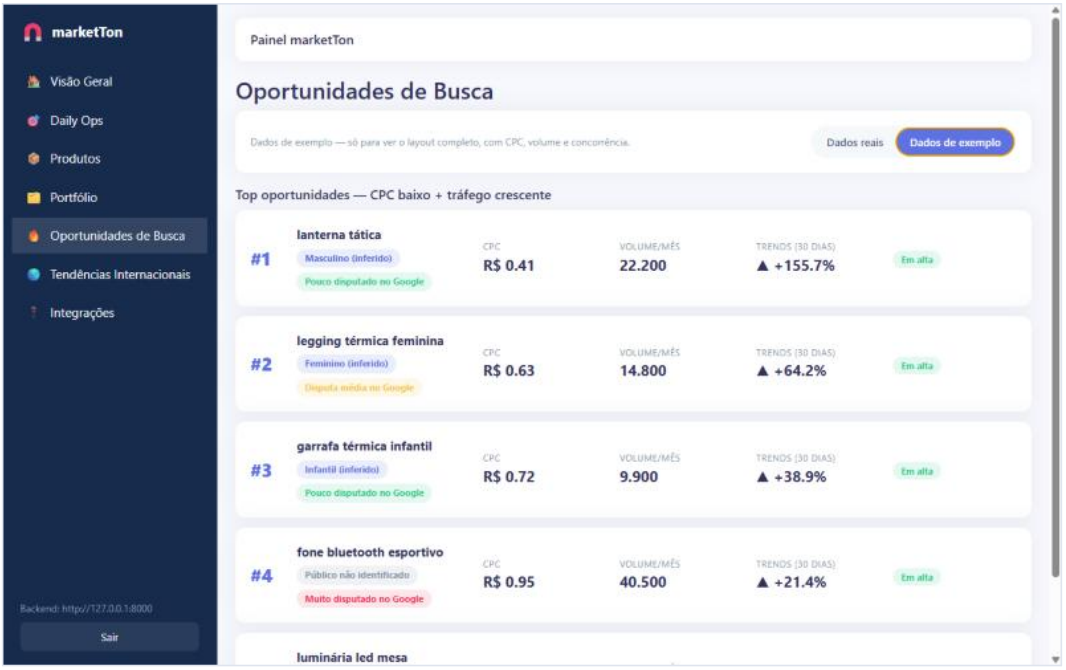



Figure 4. The same page in its 'Dados de exemplo' (sample data) mode, which is how the interface will present monthly searches, CPC and competition returned by the Google Ads API. **The values in this figure are illustrative sample data, not real Google Ads results.**

## 7. Data handling, security and privacy

- **Purpose limitation:** Google Ads data is used only to display keyword metrics to the operator inside marketTon.
- **No sharing:** Google Ads data is not sold, shared, transferred or disclosed to third parties, and is not used for advertising or to build profiles. Use of information received from Google APIs follows the Google API Services User Data Policy, including the Limited Use requirements.
- **Storage:** keyword metrics are held only in server memory for up to 6 hours and are not persisted. The only stored Google data is the OAuth refresh token and account identifiers of the operator's own account.
- **Credential protection:** credentials are stored server-side in the application database, are accepted write-only by the API and are never sent to the browser.
- **Access control:** the interface and the API require authentication; managing integrations needs a dedicated permission (connectors.manage).
- **Revocation and deletion:** the operator can revoke access at any time at myaccount.google.com/permissions and delete the stored credentials in the Integrations page.
- **Privacy policy:** https://felixsystems.com.br/privacidade.html

## 8. Compliance statement

marketTon is an internal tool that reads Keyword Planner metrics from the operator's own Google Ads account. It does not manage campaigns, does not provide advertising services to third parties, does not resell Google Ads data, and is designed to comply with the Google Ads API Terms and Conditions and is outside the Required Minimum Functionality policies applicable to tools that manage campaigns (which this tool does not). The developer token is kept private on the server and is never exposed to the browser.